David Zambrano

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EDUCATION

Florida Polytechnic University Lakeland, FL
Master of Science in Electrical Engineering. GPA: 3.5. *May 2027*
Florida Polytechnic University Lakeland, FL
Bachelor of Science in Computer Engineering. GPA: 3.88. *May 2026*
Relevant Coursework: Advanced Optoelectronics, VLSI Design, Embedded Operating Systems, Computer Architecture and Organization, Microprocessors, Object Oriented Programming, Data Structures and Algorithms, Digital Logic Design

EXPERIENCE

Undergraduate & Graduate Research Assistant May 2025 – Present
PolySilicon
- Conducting thesis research in silicon photonic integrated circuit (PIC) design, from simulation through GDSII layout and tapeout on 220 nm Silicon-on-Insulator (SOI) process.
- Completed three end-to-end tapeouts with Applied Nanotools' NanoSOI MPW service (Oct 2025, Jun 2026, Oct 2026), including a Mach-Zehnder Interferometer, novel optical ring resonator, and all-optical logic gates (NOT, NAND, NOR), operating at the center of the optical C-band (1550 nm).
- Utilized Flexcompute Tidy3D and Lumerical suite for device geometry specification and electromagnetic simulation; used KLayout, gdsfactory, and Luceda IPKISS for fabrication layout.

Electrical Systems Lead Jan 2026 – Present
Phoenix Autonomy
- Lead a six-person team within the Phoenix Autonomy research project, designing the autonomous boat's electrical system for the RoboBoat competition.
- Designed a thruster control scheme from scratch using an AM32 ESC and Raspberry Pi Pico (C++) for manual control via FlySky controller and receiver, and autonomous control with command inputs from a Jetson Orin Nano.
- Led KiCad design workshops with the undergraduate team to design the boat's main board, complete with buck converters, an RP2040 MCU, and an optocoupler-MOSFET-contactor circuit for high-power switching to thrusters.

Embedded R&D Intern May 2024 – Aug 2024
Measutronics Corporation
- Developed a PLC-based alarm system to warn dredging vehicle operators of hazardous or restricted zones outside their line of sight.
- Created custom firmware (C++) for a robust, configurable system, complete with a Windows Forms GUI (C#).

PROJECTS

Subwavelength-engineered Moire Ring Resonator Aug 2025 – Present
PolySilicon
- Designed a ring resonator with width-modulated sidewalls to induce azimuthal perturbation, enabling tunable spectra, mode splitting, and localized resonances for filtering, nonlinear optics, and optical computing.
- Electromagnetic simulations demonstrate unique disorder-induced signatures and resonances with up to 26 dB extinction ratio, suitable for hardware security applications as a Physical Unclonable Function (PUF).

Cascadable All-Optical Logic Gates May 2025 – Present
PolySilicon
- Designed an all-optical inverter from microring resonators, utilizing destructive interference effects for logic inversion in a passive configuration without electro-optic or thermo-optic tuning.
- Electromagnetic simulations confirm transmission inversion, from which 2-input NAND and NOR gates are derived; SEM imaging confirms high-fidelity geometry from a relaxed passive design.

Custom 3D Printed Nerf Blaster "Warthog" Dec 2025 – Aug 2026
Baja Blasters (Personal)
- Designed a 4-layer printed circuit board in KiCad, incorporating an RP2040 to drive an electronic speed controller for two brushless motors, with a fast decay circuit for a solenoid pusher supporting fire rates up to 25 darts per second.
- Designed a Buck-LDO circuit to supply 3.3V to low-power circuitry; wrote firmware (C++) for reconfigurable user profile settings, memory management, and an I2C OLED screen for user interaction and system feedback.

RISC-V 32I Processor Mar 2025 – May 2025
Florida Polytechnic University
- Constructed a single-core RV32I processor subset in the Intel Quartus Prime FPGA SDK, implementing 23 arithmetic and branch logic instructions.
- Validated on a DE-10 Lite FPGA board via a purpose-written assembly program to recursively calculate the Fibonacci sequence for 16 iterations.

PUBLICATIONS & PRESENTATIONS

- D. Zambrano and F. B. Tarik, "Design and Fabrication of a Subwavelength-engineered Silicon Photonic Moire Ring Resonator," CLEO 2026, Technical Digest Series (Optica Publishing Group, 2026), paper JW1.47. DOI: https://doi.org/10.1364/CLEO_AT.2026.JW1.47
- D. Brokaw, D. Zambrano, F. B. Tarik, "Scalable all-optical logic gates using cascadable silicon photonic resonator inverters," Proc. SPIE 13900, Integrated Optics: Devices, Materials, and Technologies XXX, 139000E (Mar 2026). DOI: https://doi.org/10.1117/12.3081538
- F. Gonzalez, D. Zambrano, F. B. Tarik, "Deeply sub-diffractive silicon photonic ring resonator for localized field enhancement in integrated photonics," SPIE Photonics West, 2026.
- D. Zambrano and F. B. Tarik, "Cascadable All-Optical Logic Gates in Silicon Photonics for Energy-efficient Computing and AI Acceleration," Oral Presentation, Second Annual Student Research Symposium, IEEE Student Branch, Barnett Applied Research Center (Apr 2026). First Place Paper Presentation, Florida Polytechnic University.
- D. Zambrano and F. B. Tarik, "Truly Azimuthal Superperiodic Silicon Photonic Microring Resonator for Scalable Post-Quantum Hardware Security," Poster Presentation, Second Annual Research Day, Barnett Applied Research Center (Apr 2026). First Place Outstanding Research Presentation, Florida Polytechnic University.

LEADERSHIP

- Phoenix Autonomy Electrical Systems Lead (Jan 2026 - Present)
- Phoenix Autonomy Battery Lead (Aug 2025 - Dec 2025)
- Presidential Ambassador (Jan 2023 - May 2025)
- Poly Concert Band Captain (Aug 2026 - Present)
- Poly Concert Band Vice Captain (Aug 2024 - May 2026)
- Violet Voices Choir Bass Section Lead (Aug 2024 - May 2025)

AWARDS & HONORS

- First Place Research Presentation, Second Annual Poly Research Day
- First Place Paper Presentation, Second Annual Student Research Symposium, IEEE Student Branch
- IEEE-HKN Honor Society Member
- President's & Provost's List

SKILLS

Simulation & Layout CAD: Lumerical (FDTD, MODE, INTERCONNECT), Flexcompute Tidy3D, Luceda IPKISS, KLayout, gdsfactory, Mentor Graphics Pyxis, MATLAB, Electric
EDA & Embedded Tools: KiCad, Intel Quartus Prime, STM32CubeIDE, Microchip Studio, LTspice, Multisim, SolidWorks (CSWA), Autodesk Fusion, Visual Studio Code
Programming & HDL: C++, C, C#, Python, Embedded C, Verilog, SystemVerilog, MIPS Assembly